

DS	Session	Tree 1	Question Information	Level 1	Challenge 63
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Question description

You are given an $n \times n$ grid representing the map of a forest. Each square is either empty or contains a tree. The upper-left square has coordinates $(1, 1)$, and the lower-right square has coordinates (n, n) .

Your task is to process q queries of the form: how many trees are inside a given rectangle in the forest?

Constraints

- $1 \leq n \leq 1000$
- $1 \leq q \leq 2 \cdot 10^5$
- $1 \leq y_1 \leq y_2 \leq n$
- $1 \leq x_1 \leq x_2 \leq n$

Input

The first input line has two integers n and q : the size of the forest and the number of queries.

Then, there are n lines describing the forest. Each line has n characters: $.$ is an empty square and $*$ is a tree.

Finally, there are q lines describing the queries. Each line has four integers y_1, x_1, y_2, x_2 corresponding to the corners of a rectangle.

Output

Print the number of trees inside each rectangle.

Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 4 3 .*. *.* **. *** 2 2 3 4 3 1 3 1 1 1 2 2 EXPECTED OUTPUT 3 1 2	INPUT (STDIN) 4 3 .*.* *.* **.* ***. 1 2 3 4 3 2 3 4 1 4 2 3 EXPECTED OUTPUT 6 2 0

▼ Mandatory Test Cases

Test Case 1
KEYWORD
<code>for(i=1;i<=n;i++)</code>

▼ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
8

Test Case 2
TOKEN COUNT
310

Test Case 3
NLOC
28